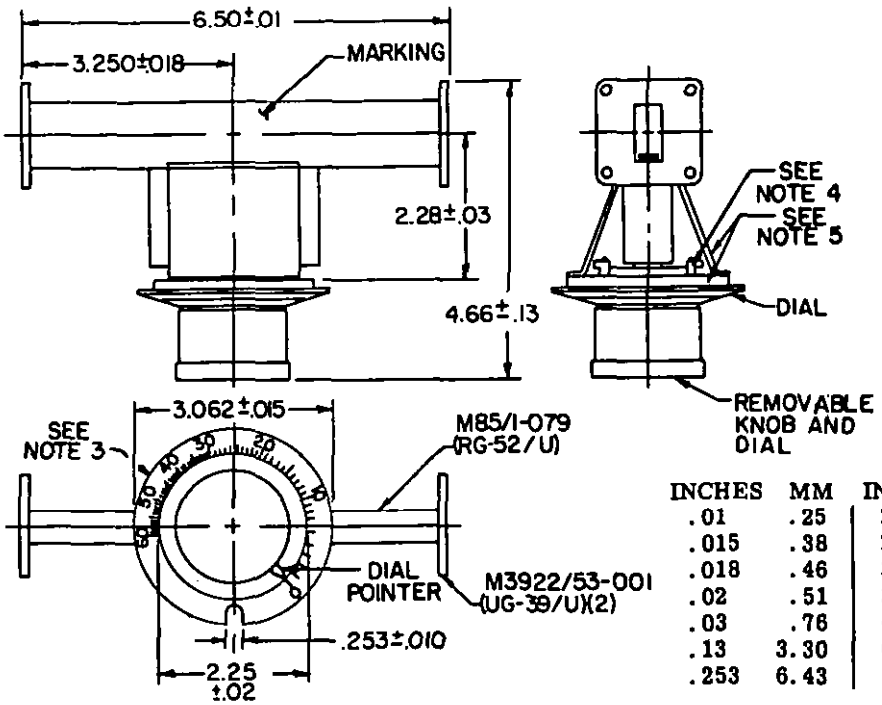
MILITARY SPECIFICATION SHEET

ATTENUATORS, VARIABLE (WAVEGUIDE)
CLASS 14

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the attenuators described herein shall consist of this document and the latest issue of Specification MIL-A-24215.



INCHES	MM	INCHES	MM
.01	.25	2.25	57.15
.015	.38	2.28	57.91
.018	.46	3.062	77.77
.02	.51	3.250	82.55
.03	.76	4.66	118.36
.13	3.30	6.50	165.10
.253	6.43		

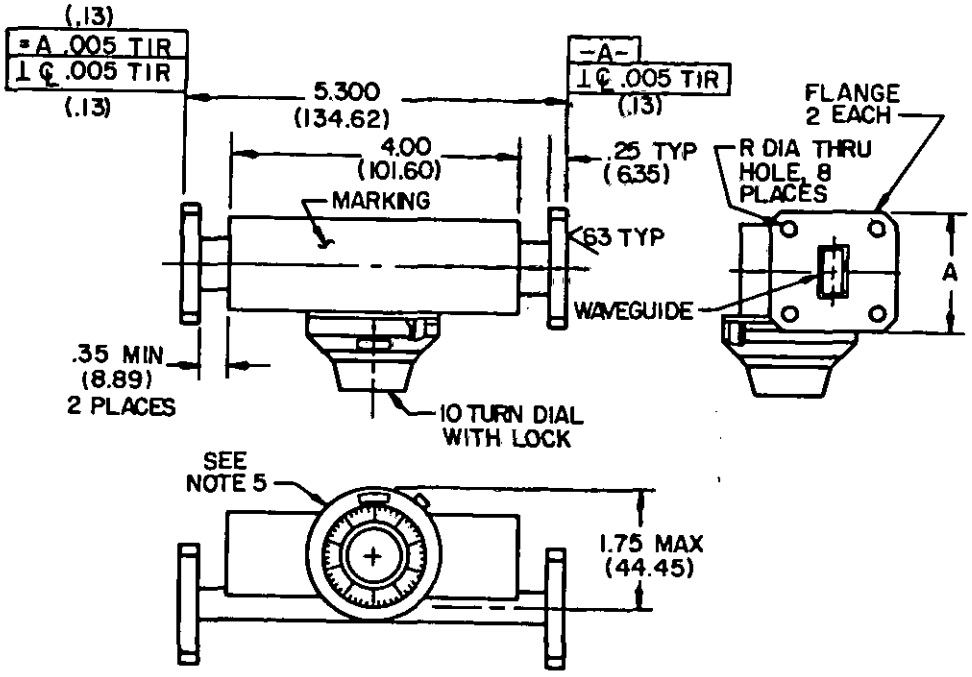
NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Dial, aluminum, painted black, with white filled engraved numerals and graduations as shown. Rotation of 230° for 0 to 60 dB. Dial pointer to be engraved and white filled. Accuracy from 0 to 50 dB, ±0.6 dB; from 50 to 60 dB, ±1.0 dB at 9 GHz.
4. Unit shall be packaged with dial assembly and suitable hardware and mounting clamp for mounting through a 1.50 (38.10 mm) diameter hole in a .12 (3.05 mm) thick panel with dial assembly secured.
5. Unit shall be mounted on a dummy panel with shipping bracket attached. As normally mounted through a panel, O-ring gaskets shall be provided between the dial and panel for a mechanical seal to prevent the entry of moisture and foreign matter through the panel opening.

FIGURE 1. Dimensions and configuration for part number M24215/4-001.

Ⓑ denotes changes.

FSC 5985

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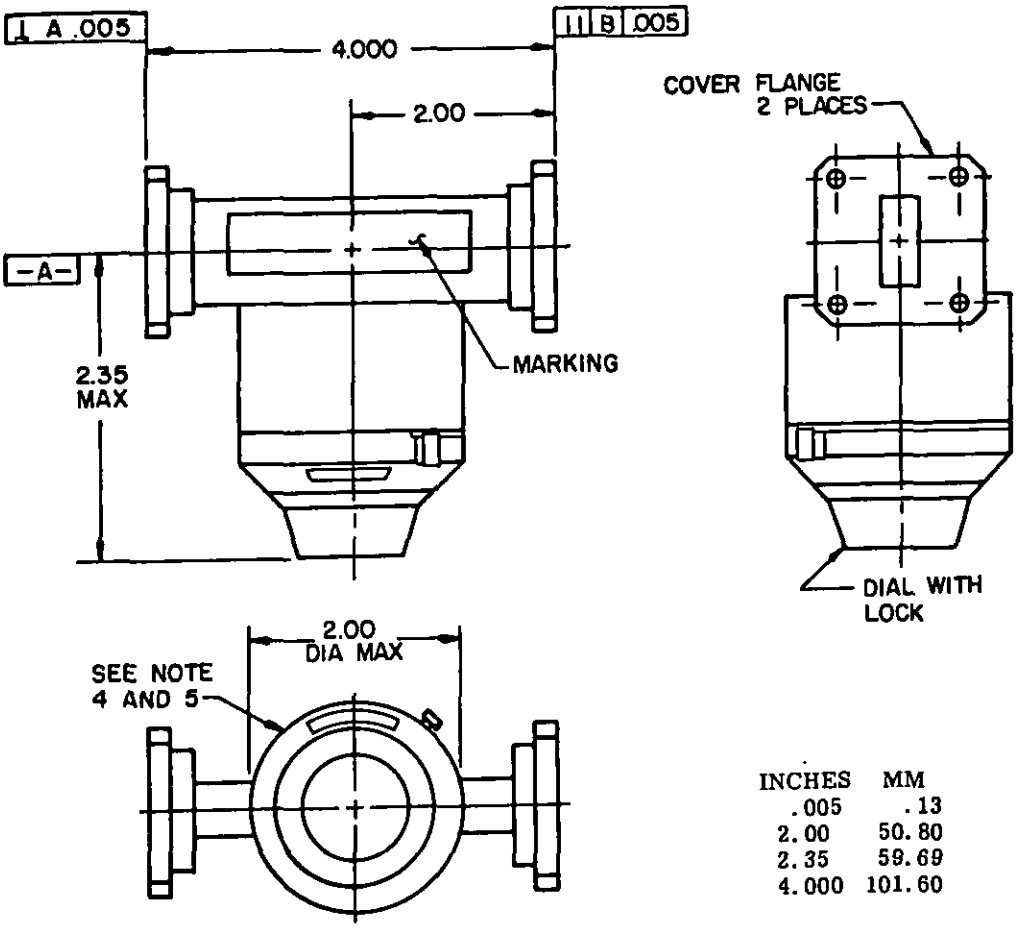


Part no. M24215/4-	A	R
002	1.69 (42.93)	.169 (4.29)
003	1.31 (33.27)	.144 (3.66)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Metric equivalents are in parentheses.
4. Unless otherwise specified, tolerances are ±.010 (.25 mm) for three place decimals and ±.03 (.76 mm) for two place decimals.
5. Dial: uncalibrated - 10 turns of dial for minimum to maximum attenuation. 100 divisions per turn of dial, numbered every 10 divisions.

Ⓑ FIGURE 2. Dimensions and configuration for part number M24215/4-002 and 003.

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INCHES	MM
.005	.13
2.00	50.80
2.35	59.69
4.000	101.60

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.010 (.25 mm) for three place decimals and ±.03 (.76 mm) for two place decimals.
4. Clockwise revolution of dial shall smoothly increase attenuation. Dial shall be marked with 100 divisions per turn, but calibration is arbitrary. Approximately 4 turns shall be required for 0 to 30 dB at 15 GHz.
5. Dial backlash: 0.5 dB maximum difference in attenuation when any dial setting is approached from either direction after cycling 500 times.

Ⓑ FIGURE 3. Dimensions and configuration for part number M24215/4-004, 005, and 006.

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(B) REQUIREMENTS:

Dimensions and configurations: See figures 1, 2, and 3.

Electrical requirements: See table I.

Part number: See figures 1, 2, and 3.

TABLE I. Electrical requirements.

Part number M24215 /4- ( )	Figure	Freq. range (GHz)	Flange	Atten. range (dB) 6/	Dial back-lash	VSWR (max)	Insertion loss (dB)	Calibration accuracy (dB)	Frequency sensitivity (dB)	Max input (watts)		Nominal operating temp. range (°C)	Weight (max)		
										av.	pk.		lbs	g	oz
001	1	8.5-9.6	M3922/53-001 (UG-39/U)	0-60	-	1.15:1	1.0	-	±2	1	-	0-65	2.5	1134	40
002 (See notes 1 and 2)	2	7.0-11.0	UG-1493 /U (See note 3)	0-20	0.1	1.25:1	0.3	-	.1 or 2%	5	100	0-65	5	2268	80
003 (See notes 1 and 2)	2	10.8-18.0	M3922/53-005 (UG-419/U) (See note 3)	0-20	0.1	1.25:1	0.3	-	.1 or 2%	5	100	0-65	4	1815	64
004 (See note 4)	3	11.0-17.0	M3922/53-006 (UG-1665/U) (See note 5)	0-30	See fig. 3	1.4:1	0.5	-	±2	1	100	0-65	.5	227	8
005 (See note 4)	3	12.4-18.0	M3922/53-006 (UG-1665/U) (See note 5)	0-30	See fig. 3	1.15:1	0.5	-	±2	1	100	0-65	.5	227	8
006 (See note 2)	3	12.4-18.0	M3922/53-005 (UG-419/U)	0-30	See fig. 3	1.15:1	0.5	-	±2	1	100	0-65	2.0	907	32

NOTES:

1. Pressurization: Leakage shall not exceed .005 cubic inches of air per minute when waveguide section is pressurized at 20 PSI.
2. Material: All parts carrying RF power (including the flanges) shall be of oxygen free high conductivity copper.
3. Except as shown on Figure 2.
4. Material: All parts carrying RF power (including the flanges) shall be of aluminum alloy.
5. Except flange is .25" (6.35 mm) thick.
6. Continuously variable.

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Ⓑ NOTE: Attenuators covered by this specification sheet are substitutable for the following commercial types:

Part number M24215/4-	FSN 5985	Manufacturer's code	Commercial type
001	821-4405	77327	196DS10
001	821-4405	18876	9176581
001	518-6245	77327	196D
001	518-6245	87864	TR641351
002		05869	723326-2
002		14135	560039
002		05869	72333-1
002		14135	560040
003		05869	723326-1
003		14135	570007
003		05869	723331-2
003		14135	570009
003		05869	721790-1
003		00443	70392-3
004		05869	723373-2
004		00443	70429
005		05869	723373-1
005		00443	711 Spec
006		00443	711

Cross reference data: Part number M24215/4-001 is interchangeable with type designation CN-1138/U.

Custodians:

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL
Navy - AS, OS
Air Force - 11, 17, 80
DSA - ES

User activities:
Army - AV
Navy - MC, CG, SH
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5985-0794)

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